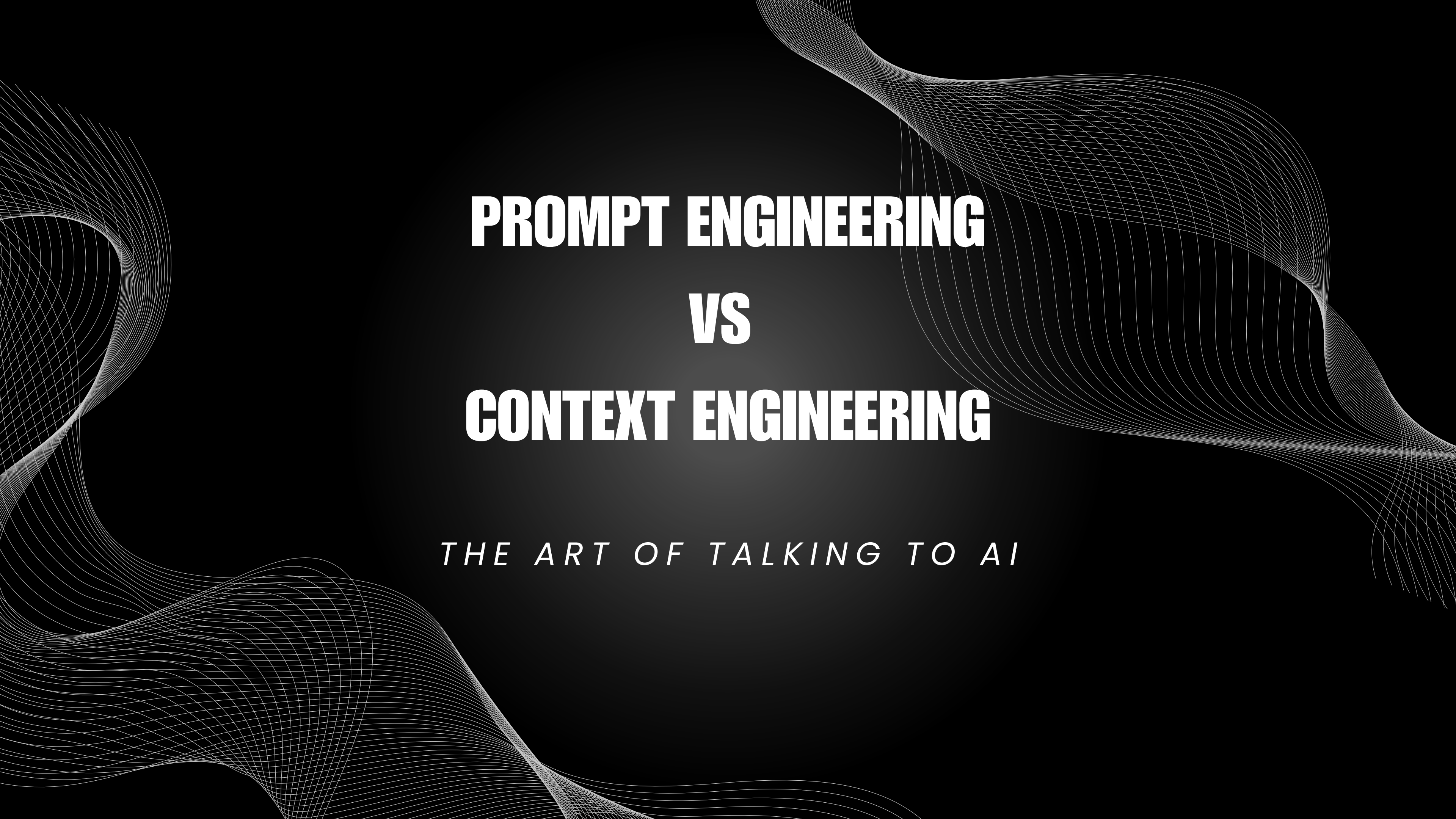


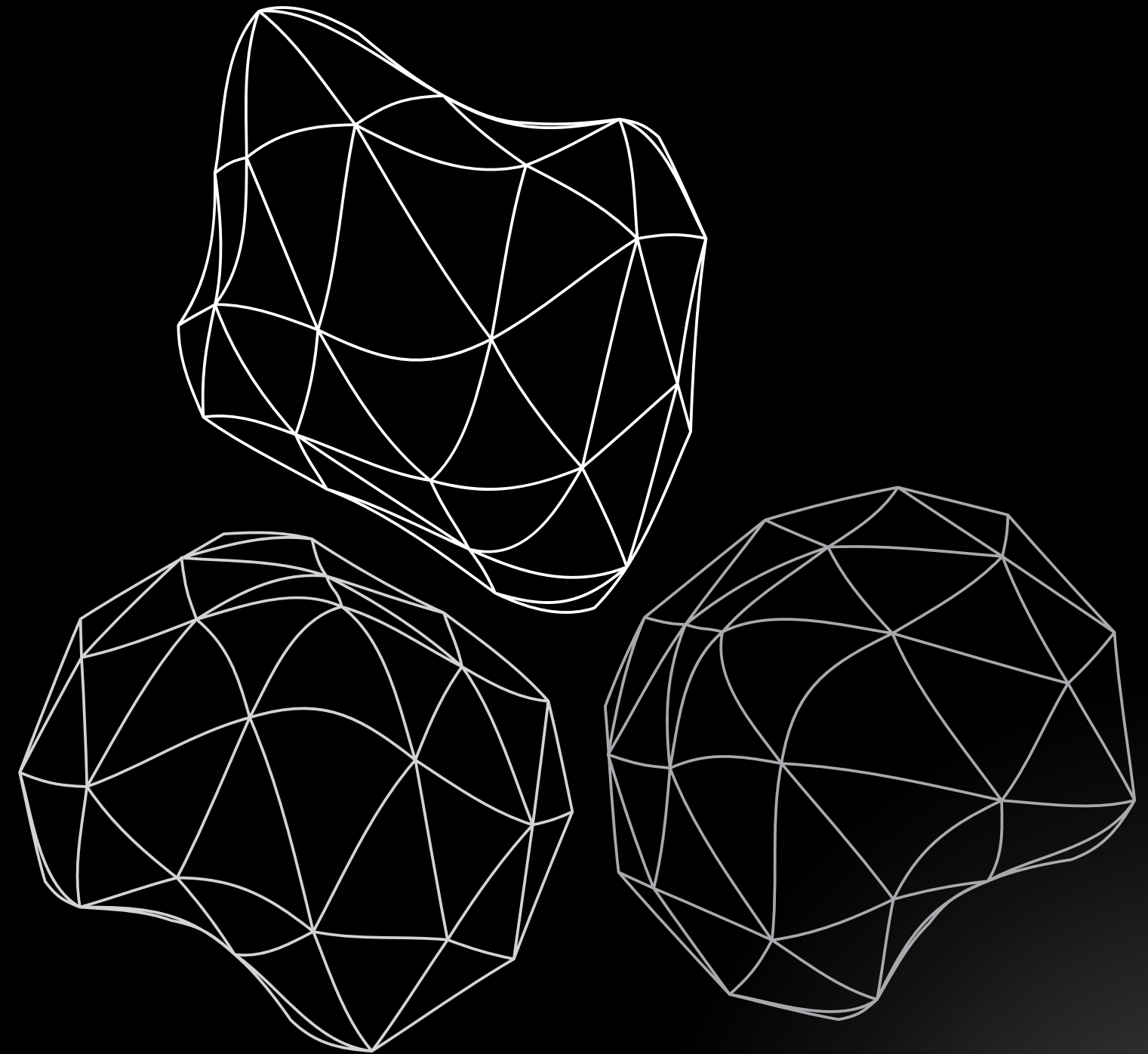


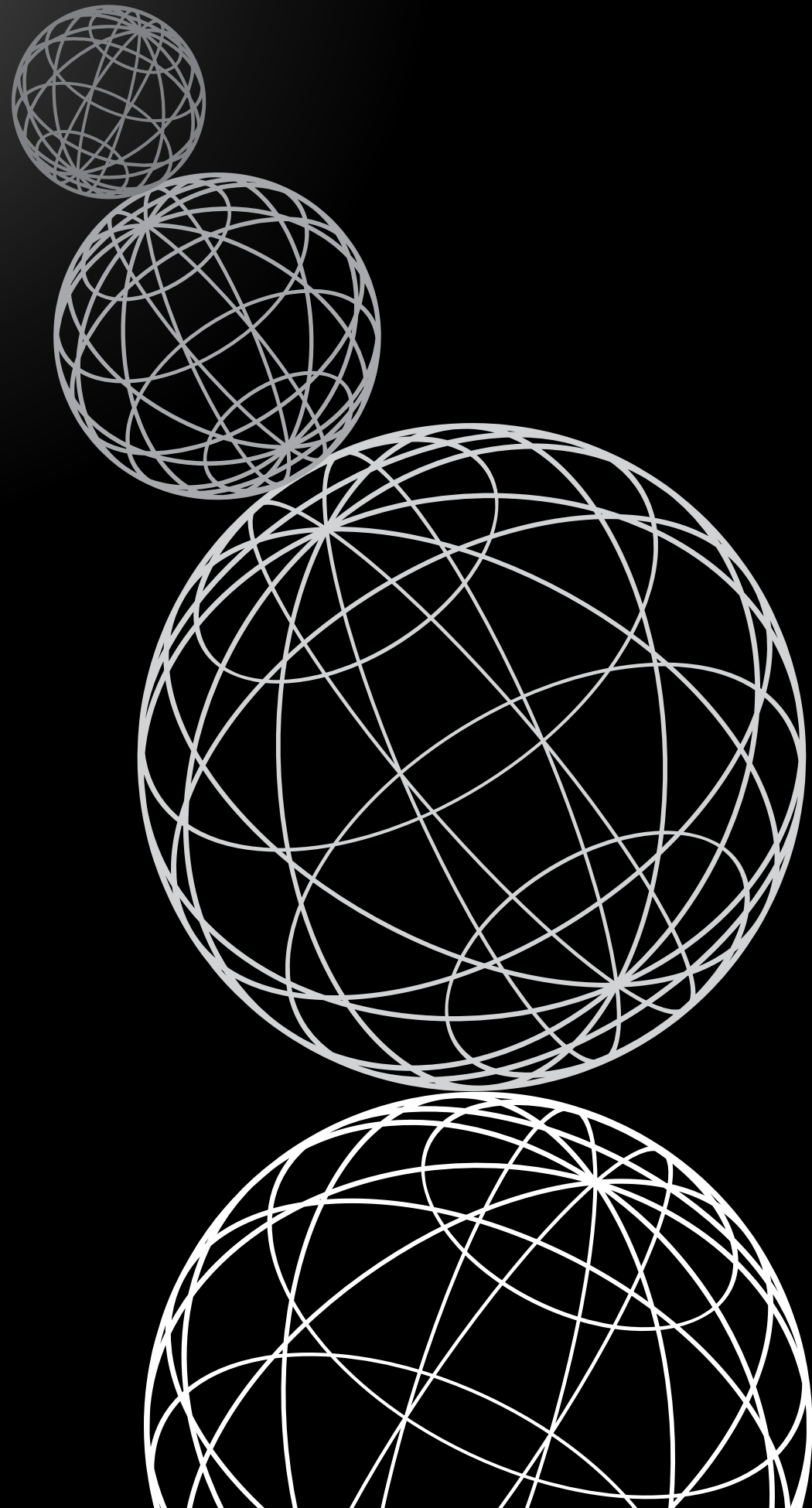
PROMPT ENGINEERING VS CONTEXT ENGINEERING

THE ART OF TALKING TO AI

WHY IT MATTERS

- *AI doesn't "know" anything — it interprets text inputs.*
- *Our skill to communicate clearly with AI defines the quality of its output.*
- *Prompt and Context Engineering = bridge between human intent & AI reasoning.*





WHAT IS PROMPT ENGINEERING ?

- *The practice of designing effective instructions or questions for AI models.*
- *Prompts are like programming — but in natural language.*
- *A strong prompt = accurate, relevant, and creative AI responses.*

Example:

- 🖱️ *“Write a professional email” vs “Write a polite follow-up email for a delayed meeting reminder.”*

KEY ELEMENTS OF A GOOD PROMPT



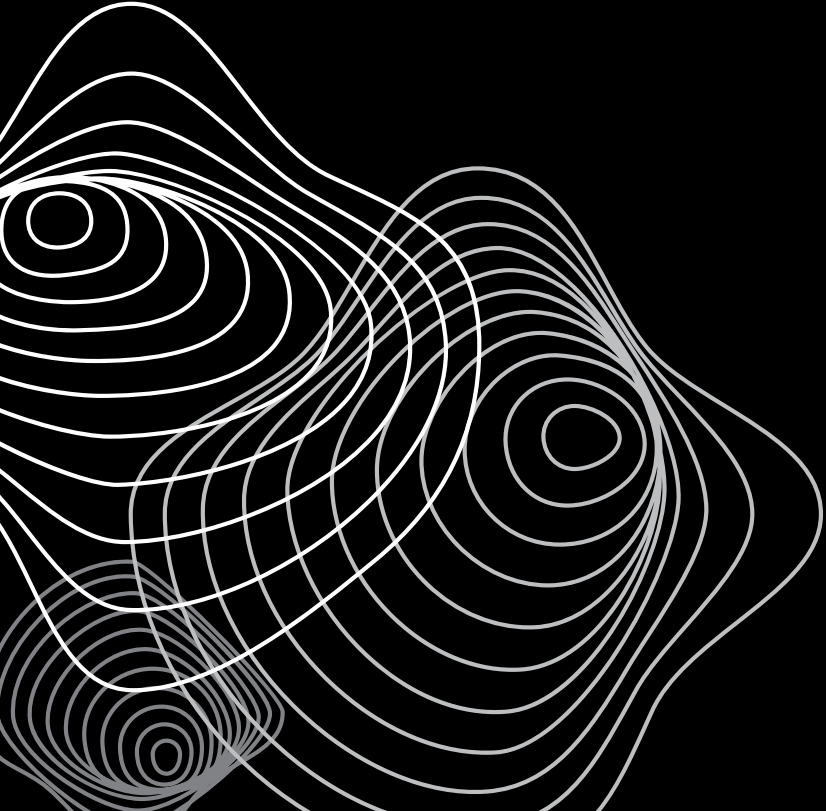
Role specification: Define the AI's role (e.g., "You are a data analyst").

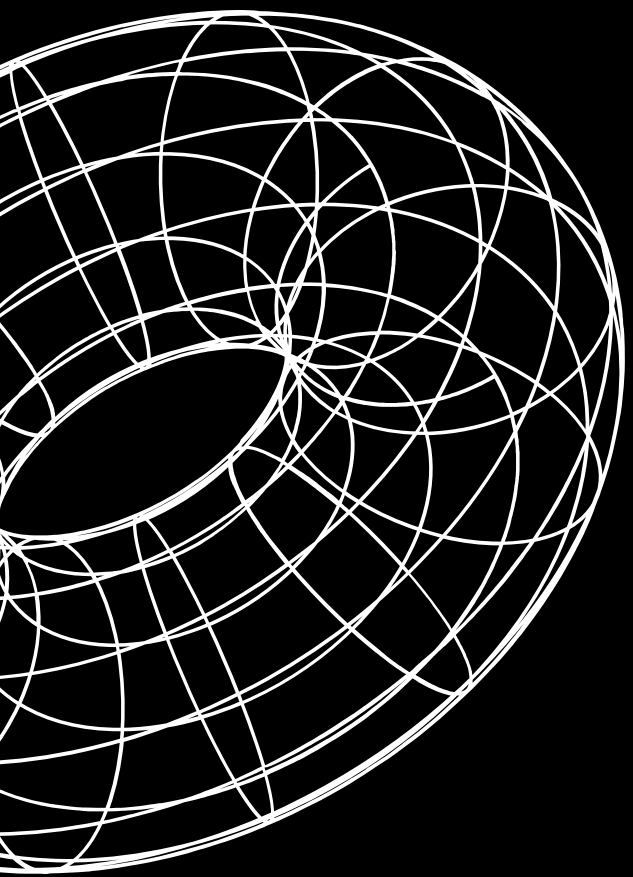
Task clarity: Explain what you need.

Constraints: Add rules (e.g., "limit to 150 words").

Examples: Few-shot prompting to show expected pattern.

Iteration: Refine based on AI feedback.

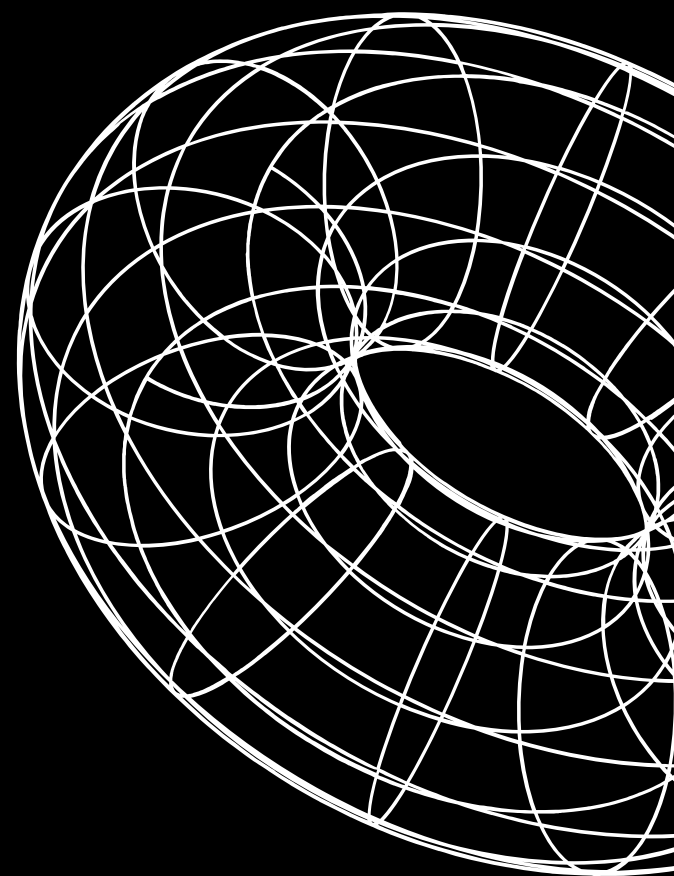




WHAT IS CONTEXT ENGINEERING?

Goes beyond prompts — focuses on what AI knows while answering.

*Context = memory + reference + background knowledge.
Helps AI connect ideas across documents, chats, or history.*





COMPONENTS OF CONTEXT

Documents

*Attached text, datasets,
or uploaded files.*

Memory & History

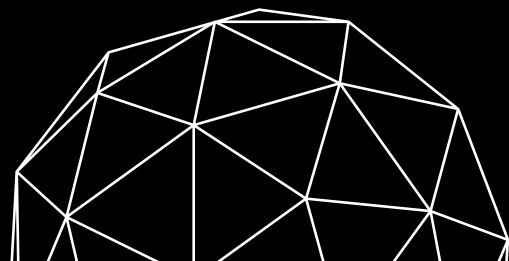
*Chat history, user
preferences, session
context.*

Knowledge Graphs

*Linked facts and
relationships.*

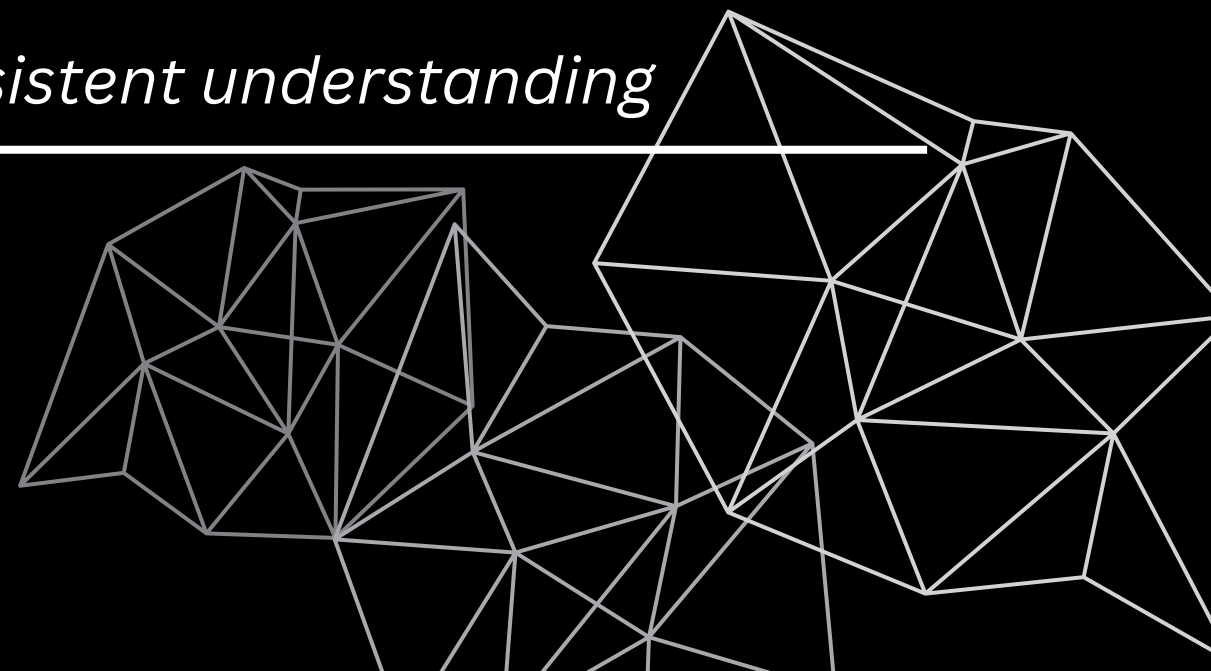
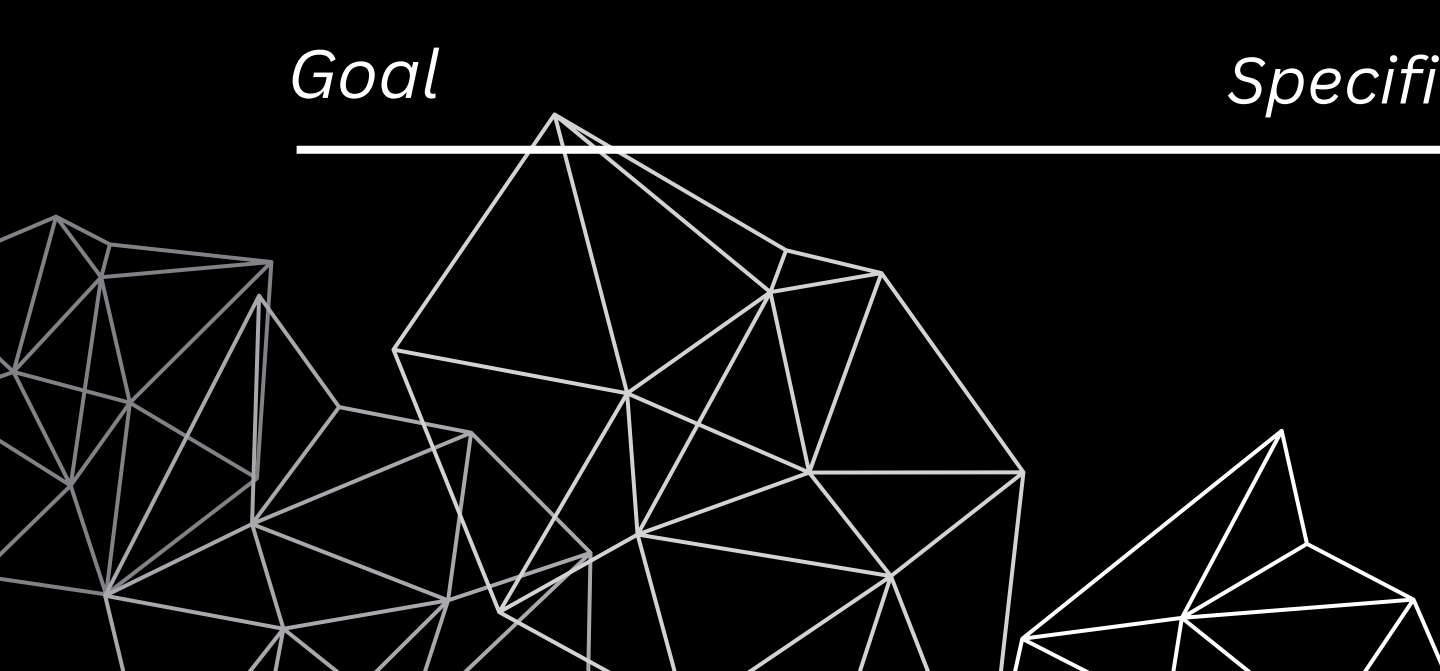
Tools & Outputs

*API calls, embeddings,
vector databases, etc.*



PROMPT VS CONTEXT — KEY DISTINCTIONS

Feature	Prompt Engineering	Context Engineering
Focus	Wording the input	Managing background knowledge
Scope	One-time instruction	Continuous learning & recall
Role	Directs model	Grounds model
Goal	Specific response	Consistent understanding



VISUALIZATION — HOW THEY INTERACT

Human → Prompt (Instruction) → AI Model

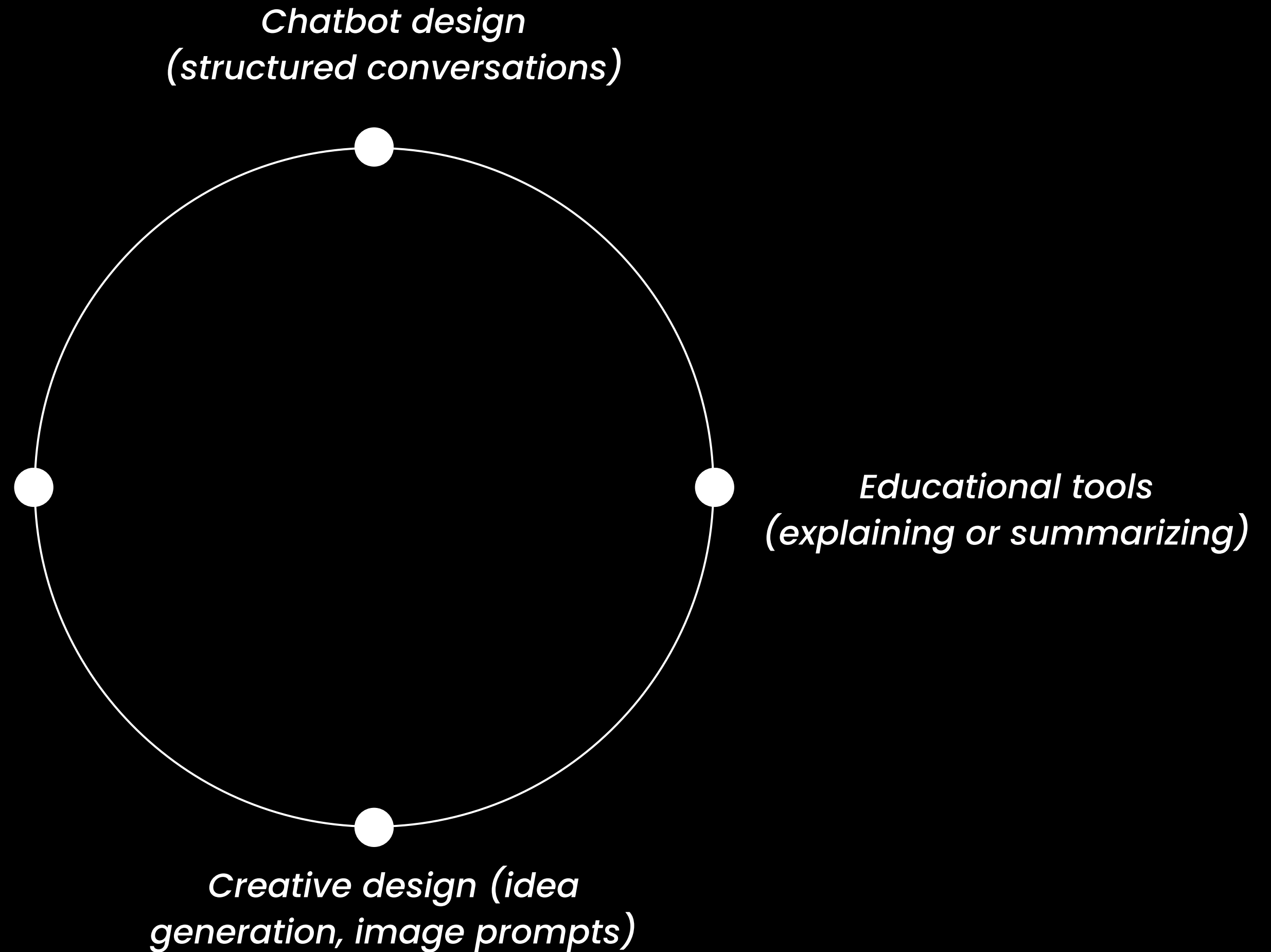


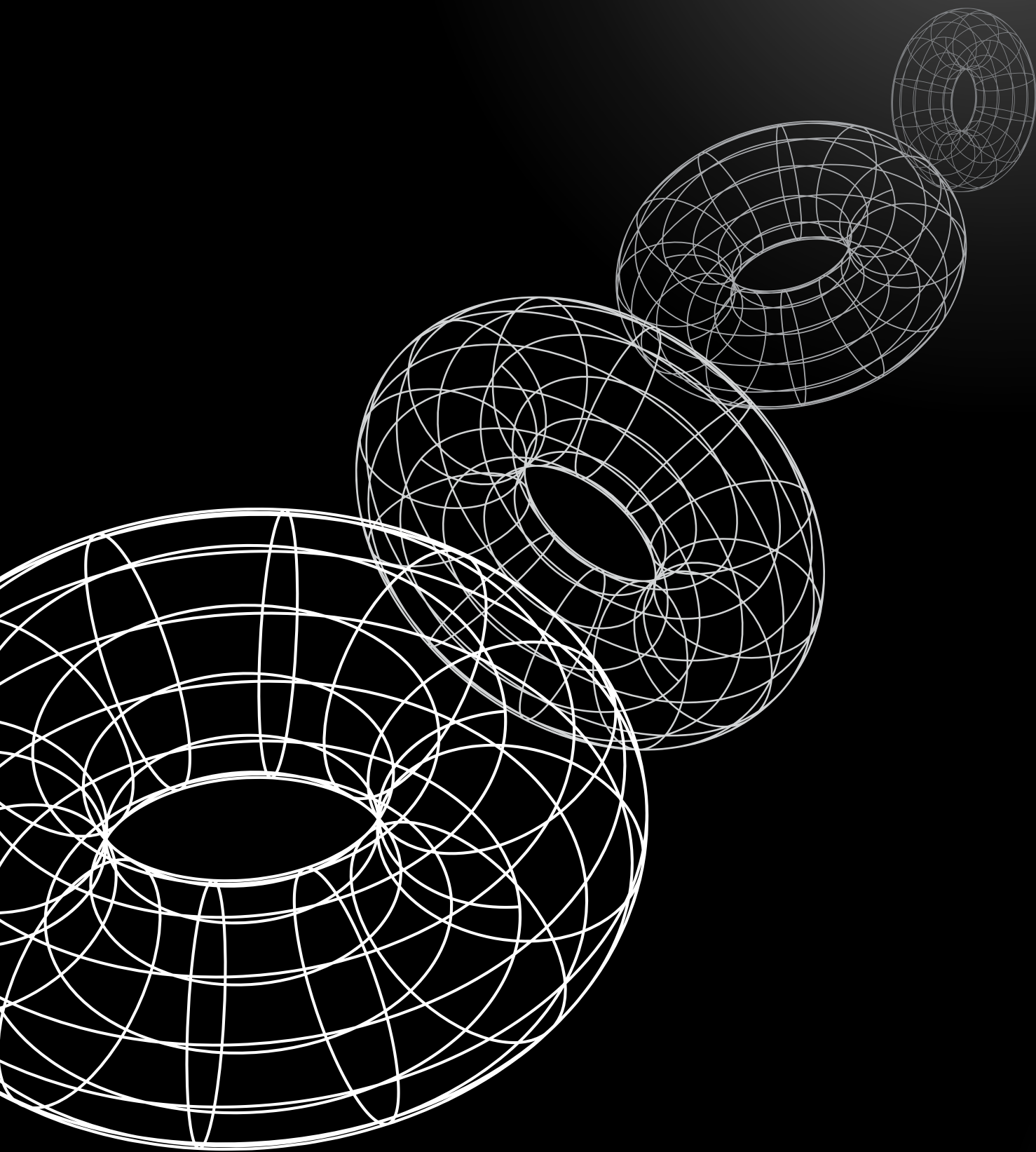
Context (Documents, Memory, Tools) → Enhanced Output

AI combines prompt intent + contextual knowledge = accurate answer.

Real-World Use Cases

Prompt Engineering

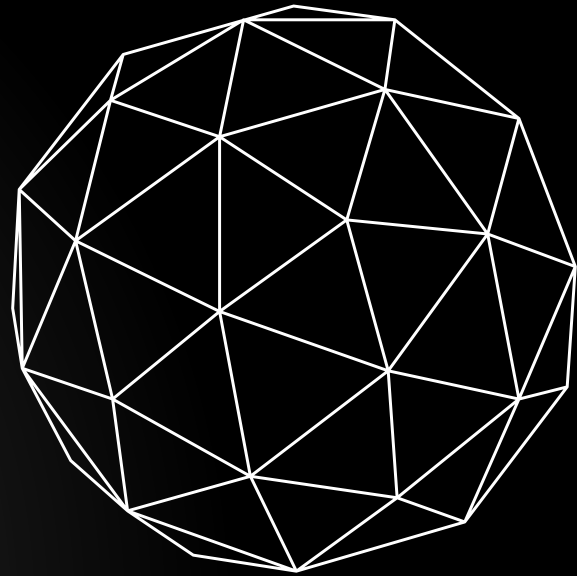




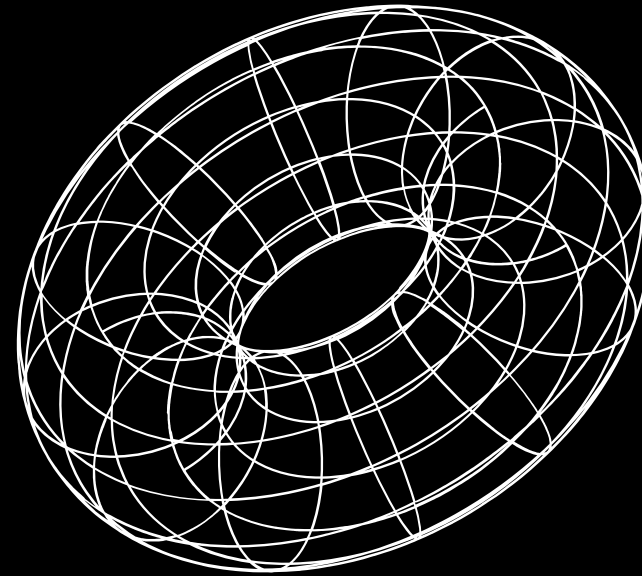
REAL-WORLD USE CASES — CONTEXT ENGINEERING

- *Customer support chatbots with company data.*
- *Legal or medical assistants accessing large document sets.*
- *Personal AI agents remembering user preferences.*
- *AI code assistants using project repositories as context.*

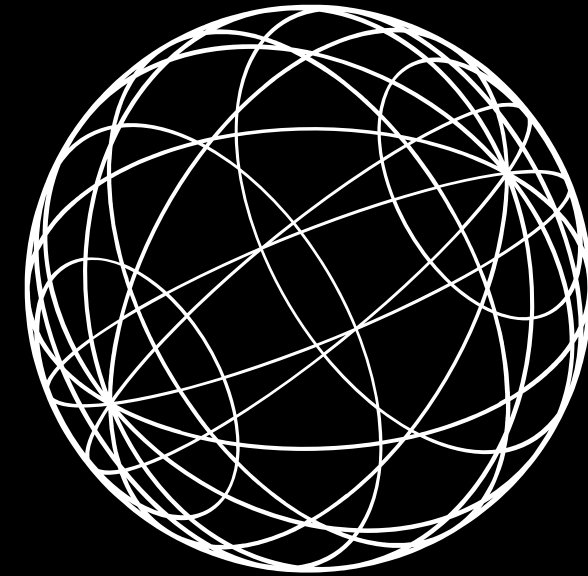
CONSEQUENCES OF POOR ENGINEERING



*Unclear prompts →
wrong or vague answers*

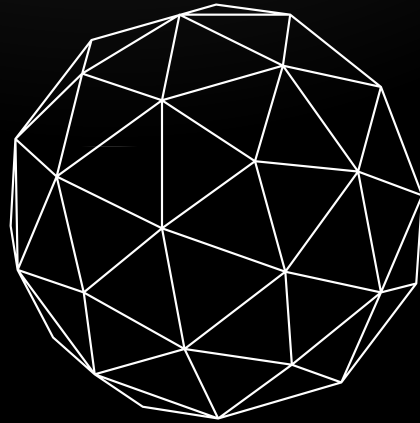


*Missing context → hallucinations
or misinformation*

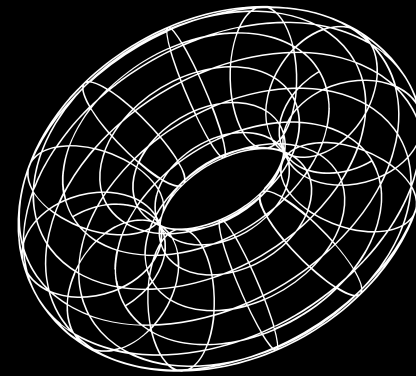


*Poor structure → wasted
tokens & weak results*

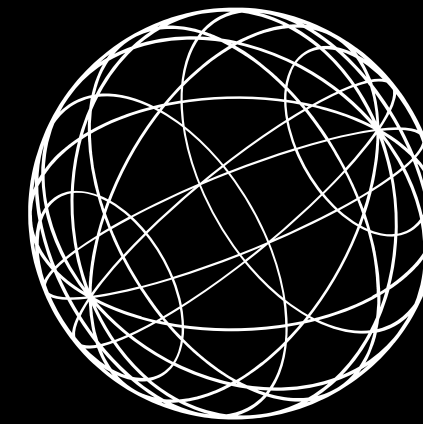
WHY CONTEXT ENGINEERING MATTERS MORE TODAY



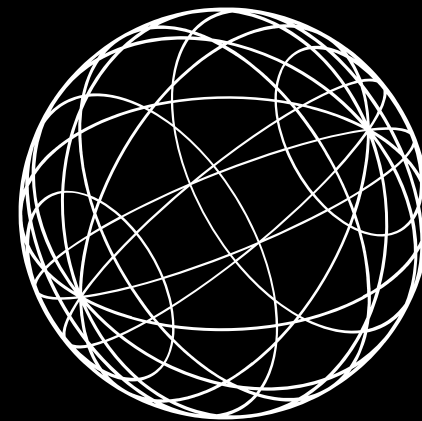
*Modern LLMs have token limits
— they can't read everything*



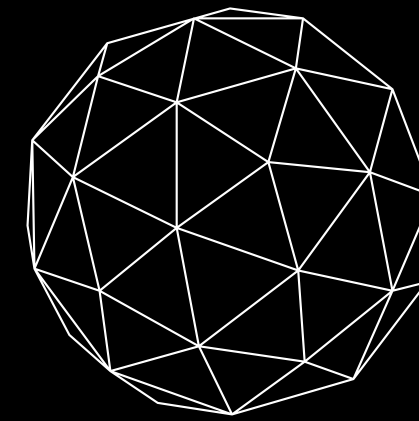
*Attention constraints: Models
forget old info in long chats*



*Context Rot: Over time,
irrelevant info crowds useful
memory*

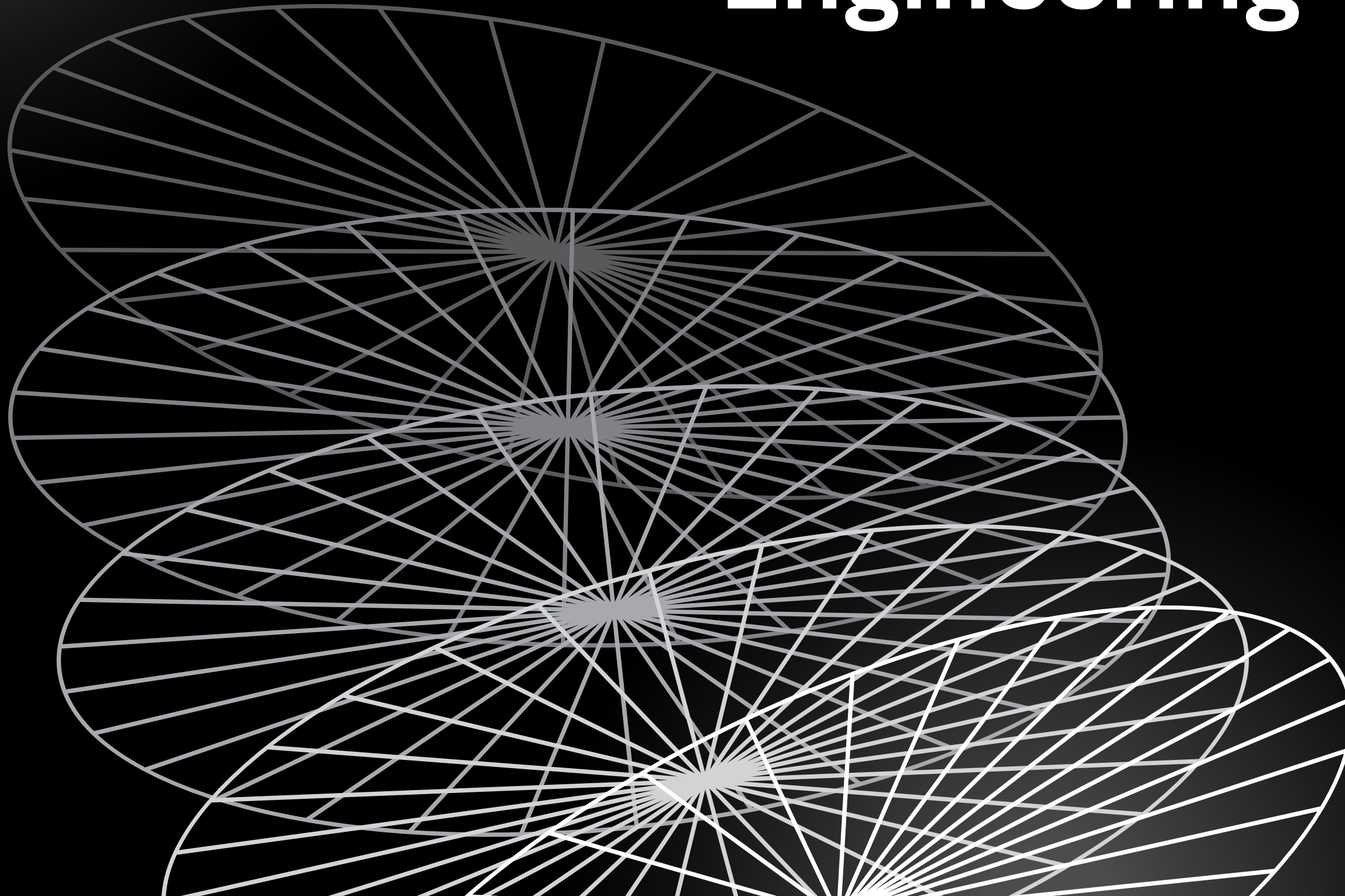


*Strategic Information
Architecture: Smart chunking,
retrieval & summarization
improve efficiency*



*Scalability: Enables multi-agent
systems & long-term
personalization*

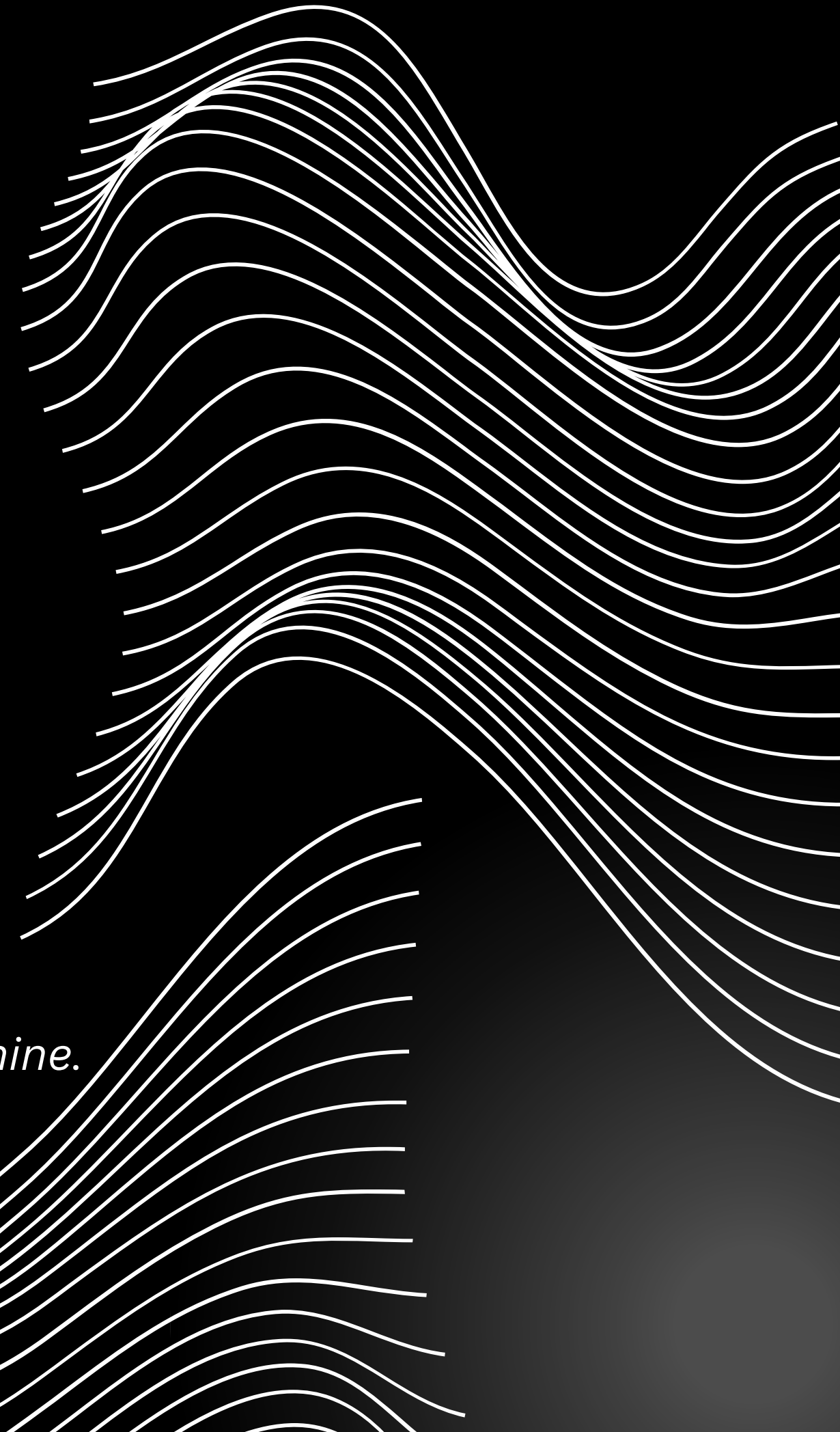
Tools Supporting Context Engineering



- *Vector Databases: Pinecone, Chroma, Weaviate*
- *Retrieval-Augmented Generation (RAG): Combines documents with prompts.*
- *Memory APIs: ChatGPT Memory, LangChain Memory.*
- *Embeddings: Convert text into contextual meaning.*

FROM PROMPTING TO CONTEXTUAL THINKING

- *The future of AI is context-driven reasoning.*
- *Instead of just “telling” AI, we’ll teach it what to remember.*
- *Prompt = question*
- *Context = background*
- *Together they form intelligent collaboration between human and machine.*





THANK YOU

PRESENTED BY
AIMAN KHAN